

Tnode Class Documentation

Source File: 'Tnode.h'
Namespace: dsl
Class Header: template <class T> class Tnode

Overview

The *Tnode* class is a generic bi-directional linked tree node.

Constructors

- Tnode() (default constructor)
 - **Purpose:** Assigns null to its links and the default value to its data.
- Tnode(const T& obj)
 - **Purpose:** Assigns *obj* to its data and null to its links.
 - **Parameter(s):**
 - * *lhs*: Constant *T* reference object.
- Tnode(const Tnode<T>& obj) (copy constructor)
 - **Purpose:** Copies the data of *obj* and assigns null to its links.
 - **Parameter(s):**
 - * *obj*: Constant *Tnode* reference object.

Assignment Operators

- operator=(const Tnode<T>& rhs)
 - **Purpose:** Copies the data of *rhs* only.
 - **Parameter(s):**
 - * *rhs*: Constant *Tnode* reference object.
 - **Return:** *this.

Destructor

- ~Tnode()
 - **Purpose:** Does nothing.

Methods

- data() const
data()
 - **Purpose:** Provides access to the data field.
 - **Return:** *T* reference object.
- left() const
left()
 - **Purpose:** Provides access to the left link field.
 - **Return:** *Tnode* pointer reference object.
- right() const
right()
 - **Purpose:** Provides access to the right link field.
 - **Return:** *Tnode* pointer reference object.
- p() const
p()
 - **Purpose:** Provides access to the parent link field.
 - **Return:** *Tnode* pointer reference object.

Non-Member Functions

- `clone(Tnode<T>* obj, Tnode<T>* p)`
 - **Purpose:** Returns a copy of *obj* with *p* as its parent.
 - **Parameter(s):**
 - * *obj*: *Tnode* pointer object.
 - * *p*: *Tnode* pointer object.
 - **Return:** *Tnode* pointer object.
- `erase(Tnode<T>*& obj)`
 - **Purpose:** Deletes the linked tree pointed to by *obj*.
 - **Parameter(s):**
 - * *obj*: *Tnode* pointer reference object.